

Mushroom



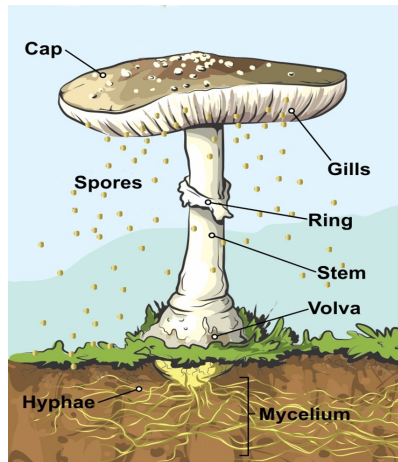
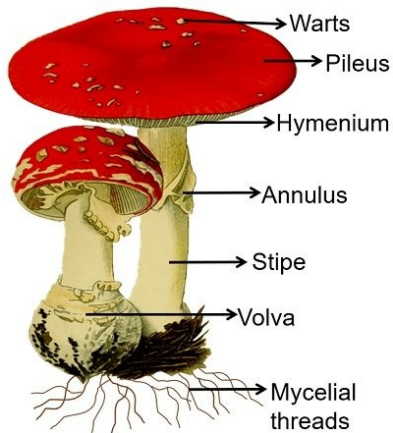
What are Mushrooms?

- A type of fungus.
- Chlorophyll is absent (achlorophyllous).
- Grows in moist and shady places: forest floor, rotting tree trunks, dung, etc.
- They are called saprophytic because they obtain their food from dead and decaying organic matter.

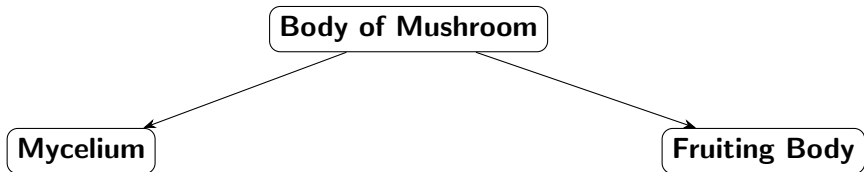
Structure of Mushroom



Structure of Mushroom



Structure of a Mushroom



Vegetative part

- Under the soil.
- Network of fine thread like fiber called hyphae.
- Absorbs water and organic nutrients.

Reproductive structure

- Above the ground.
- Parts: pileus, stipe/stalk, ring, and volva.
- Lower part of pileus contains structure known as gills.
- Gills produce spores for reproduction.

Gills

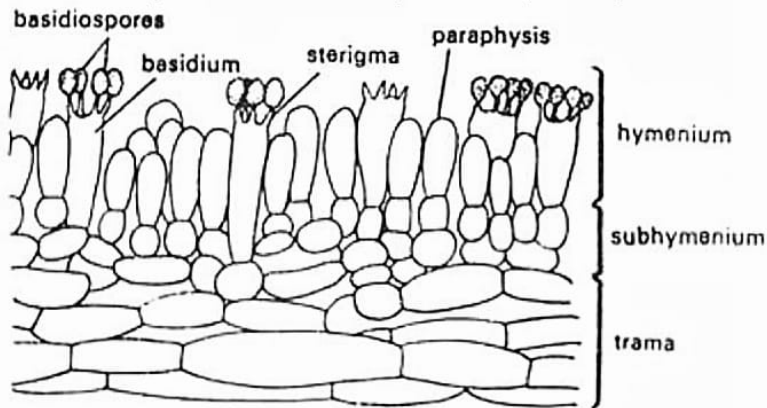
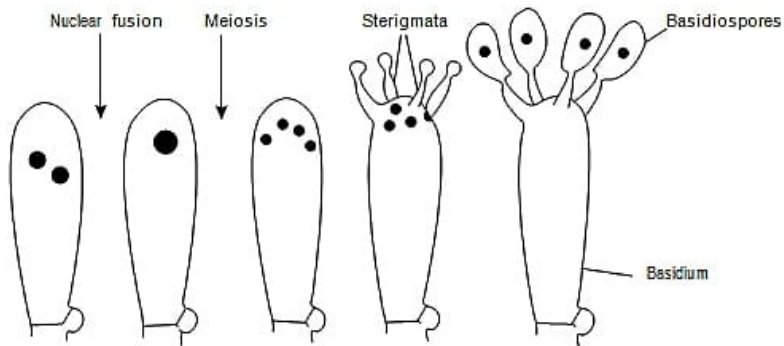


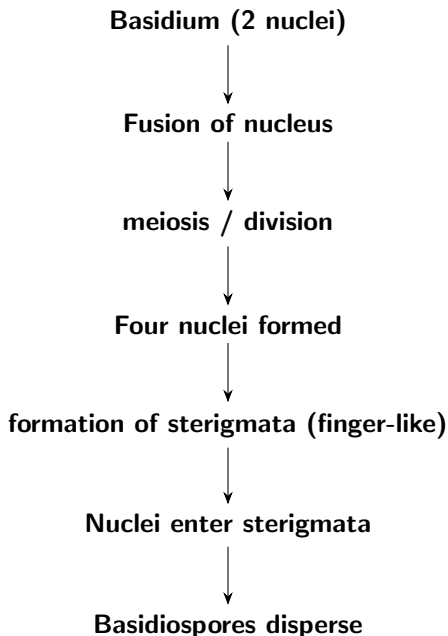
Figure: Internal structure of gills

- Three layers: trama, sub-hymenium and hymenium.
- Hymenium: Contains two types of cells- basidium and paraphysis.
- Basidium: Fertile cells → produce basidiospores for reproduction.
- Paraphysis: Sterile cells (protection/support).

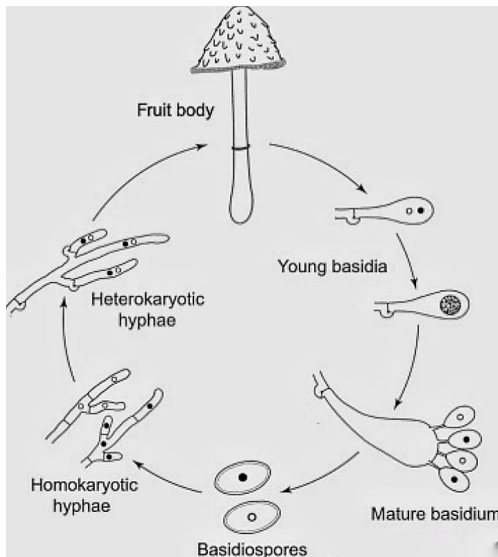
Formation of basidiospore



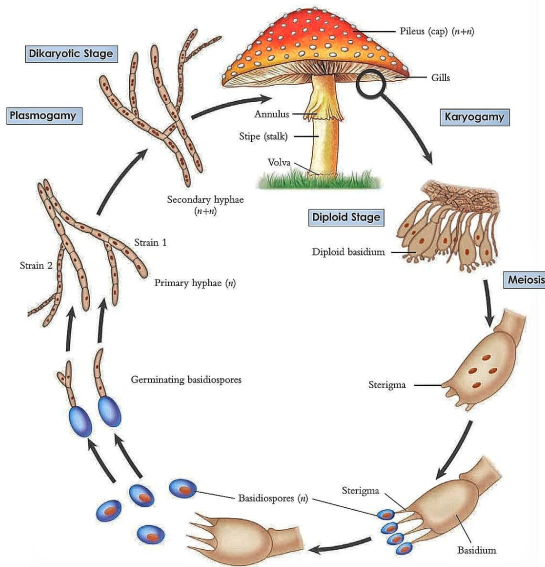
Formation of Basidiospore



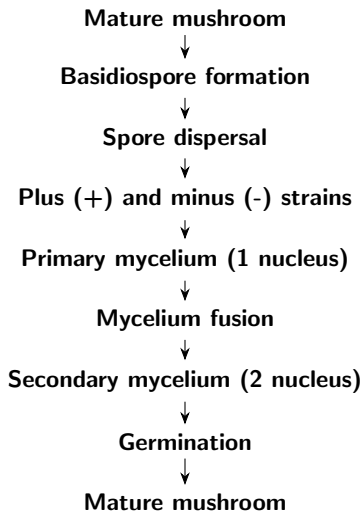
Life Cycle of Mushroom



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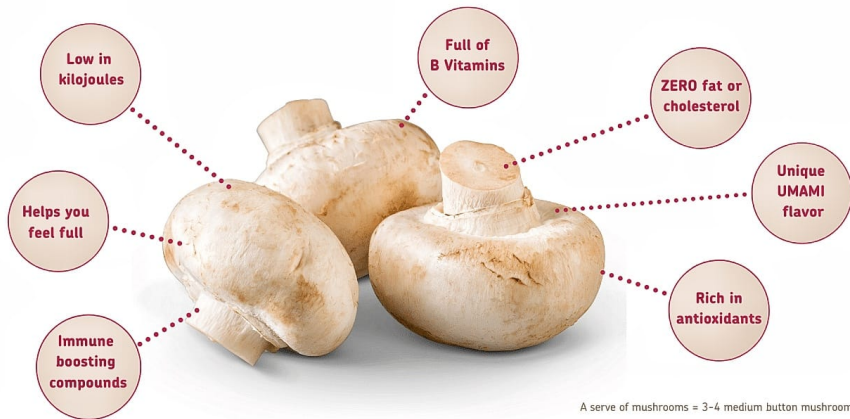


Nutrients in mushroom



Protein, Vitamins and Minerals

Nutrients in mushroom



Nutrients in Mushroom

- Good source of protein (20–35% dry weight)- growth and repair.
- Rich in vitamins: vitamin B1 (thiamine), B2 (riboflavin), B3 (niacin), B12, etc.- metabolism, skin health, blood formation, nervous system, etc.
- As a source of minerals: potassium, sodium, phosphorus, calcium, copper, iron, etc.- fluid balance, strengthens bones and muscles.

Medicinal Value in Mushroom



Figure: *Pycnoporus cinnabarinus* (Red Mushroom)

- Helps in management of diabetes, high blood pressure, heart disease and supports immunity.
- Shows potential anti-cancer and anti-inflammatory effects; used in respiratory diseases, asthma, and piles.
- Pycnoporus cinnabarinus (red mushroom) is also studied for possible use in cancer-related medicine.

Poisonous Mushroom



Amanita



Galerina



Mycena



Paxillus

Poisonous Mushroom

- Bright attractive colour.
- Volva at the base.
- White gills and white spores.
- Bad smell / bitter taste.

Economic Importance of Mushroom



Figure: Dalle (Gobre) chyau



Figure: Kanye chyau



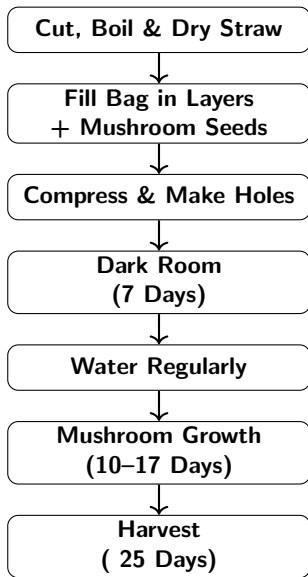
Figure: Guchi chyau

Economic Importance of Mushroom



Mushroom Farmers

Method of Kanye Chyau Farming



Mushroom Products



Sukuti



Pickle



Soup

